How-To Library

Informatica

# High availability and disaster recovery in a Data Engineering domain

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# Abstract

Learn how the Informatica domain and application services in Data Engineering meet disaster recovery and high availability requirements.

# Supported Versions

- Data Engineering 10.4.1

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# Domain overview

An Informatica domain is a collection of nodes and services.

A node is the logical representation of a machine in a domain. When you install the Informatica domain services for the first time, you designate a machine as the master gateway node where you create the Informatica domain. If you

decide to install the Informatica domain services on multiple machines, you can designate the additional machines as gateway nodes or worker nodes that you join to the domain. One of the nodes in the domain connects to the domain configuration repository database, a relational database that stores configuration and user information for the domain.

A domain also includes application services. An application service represents server-based functionality that supports the features in your Data Engineering product.

# Domain components

Include the following domain components in your high availability plan and disaster recovery strategy:

**Domain nodes**

The domain has the following types of nodes:

- Master gateway node. One machine in the domain is the master gateway node which is the entry point to the domain.
- Gateway nodes. Additional nodes can provide entry points to the domain.
- Worker nodes. Nodes that run application services but do not provide entry points to the domain.

Any node in the domain can run application services.

**Service Manager**

The Service Manager is a service that manages all domain operations. The Service Manager runs on each node in the domain and performs domain functions, such as authentication, logging and application service management.

The Service Manager on a gateway node performs more tasks than the Service Manager on a worker node.

**Application services**

Application services represent server-based functionality that support the features in your Data Engineering product.

**Databases**

Some application services require databases to store metadata and to write run-time results.

**Informatica client tools**

Informatica clients access underlying Informatica functionality in the domain. The clients make requests to the Service Manager and to application services.

# Domain nodes

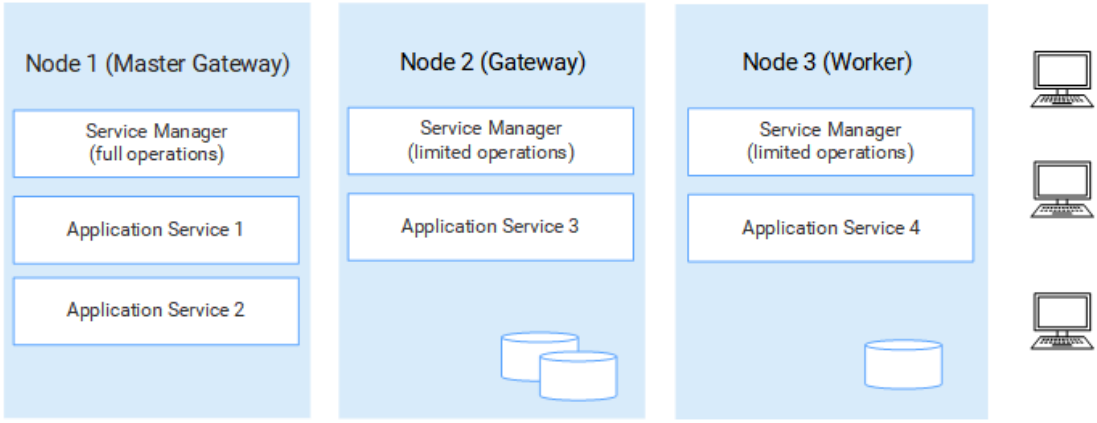
A domain includes one master gateway node and any number of gateway nodes and worker nodes.

The domain includes the following types of nodes:

- Master gateway node. One machine in the domain is the master gateway node which is the entry point to the domain. The master gateway node runs management tasks, runs the Administator tool, updates the domain configuration repository database, and can run application services.
- Gateway nodes. Additional nodes that can provide entry points to the domain and run application services. A gateway node can be promoted to the master gateway node and begin running management tasks, running the Administrator tool, and updating the domain configuration repository database.
- Worker nodes. Nodes that run application services but do not provide entry points to the domain or update the domain configuration repository database.

You can change a worker node to a gateway node and you can change a gateway node to a worker node.

The following image shows a high-level architecture of a domain on multiple nodes:




Each node in the domain runs a Service Manager process and any node can run application services. All nodes connect to the master gateway node.

## *High availability for domain nodes*

High availability functionality helps minimize service downtime due to unexpected failures or scheduled maintenance in the Informatica environment. It also eliminates a single point of failure in a domain.

High availability consists of the following components:

- Resilience. An Informatica domain can tolerate temporary connection failures until either the resilience timeout expires or the failure is fixed.
- Restart and failover. A process can restart on the same node or on a backup node after the process becomes unavailable.
- Recovery. Operations can complete after a service on a node is interrupted. After a service process restarts or fails over, it restores the service state and recovers operations.

Domain nodes are highly available and have fallback mechanisms based on the number and types of nodes in the domain:

**Domain with one node**

In a domain with one node, the domain goes down when the node goes down. Any services that are running on the node will fail.

**Domain with two nodes: a worker node and a gateway node**

In a domain with one worker node and one gateway node, the following situations can occur:

- If the gateway node fails, the worker node and all application services go down.
- If the worker node fails, the gateway node and the domain continue running. Application services on the failed worker node can fail over to the gateway node.

**Domain with two gateway nodes**

In a domain with two gateway nodes, the following situations can occur:

- If the master gateway node fails, the other gateway node becomes the master gateway node and the domain continues running. Application services on the failed master gateway node fail over to the new master gateway node.
- If the non-master gateway node fails, the application services on the failed gateway node fail over to the master gateway node.

**Multi-node domain with more than two gateway nodes**

In a domain with more than two gateway nodes, the following situations can occur:

- If the master gateway node fails, a different gateway node becomes the master gateway node and the domain continues running.
- If any other node fails, the application services fail over to another node in the domain and continue running.

### *Disaster recovery for domain nodes*

The domain relies on the domain configuration repository database to run. To prepare for disaster recovery, replicate the database across multiple data centers.

For example, consider a domain with gateway nodes that span two data centers where the master gateway node is in the first data center. The domain configuration repository database is replicated across both data centers and both copies of the database point to the first data center.

If the first data center goes down, the gateway node in the second data center is promoted to be the master gateway node and it can connect to the replicated database the second data center. To connect the master gateway node in the second data center to the replicated database, you can modify the values of the database host to point to the second data center host, or you can set up a load balancer in advance.

Because the master gateway node provides the Administrator tool web interface, you can also set up a load balancer to run the Administrator tool across the gateway nodes.

## Model Repository Service

The Model Repository Service is an application service that manages the Model repository. The Model repository stores metadata for Data Engineering products in the Model repository, a relational database that enables collaboration among the products.

The following services and tools store metadata in the Model repository:

- Data Integration Service
- Administrator tool
- Analyst tool
- Developer tool

The Model Repository Service connects to the Model repository when you run a mapping, mapping specification, profile, or workflow.

### *High availability for the Model Repository Service*

Informatica clients that connect to the Model Repository Service are resilient to temporary connection failures during failover and restart of the service.

Informatica clients can reconnect to the Model Repository Service after a temporary connection failure. If the connection is not restored before the resiliency timeout and the number of retries to reconnect to the service, the Model Repository Service becomes unavailable. The Service Manager restarts the Model Repository Service on the same node or on a backup node.

The Model Repository Service fails over to a backup node in the following situations:

- The Model Repository Service fails and the primary node is not available.
- The Model Repository Service is running on a node that fails.

### *Disaster recovery for the Model Repository Service*

Prepare for disaster recovery by configuring a backup Model Repository Service.

Complete the following tasks:

- Configure the Model Repository Service with a backup on another node.
- Back up the Model Repository Service to save repository objects and the search index in a backup file.
- Replicate the Model repository database across multiple data centers.
- Set up a load balancer to store metadata in a backup database if the primary database fails.

## Data Integration Service

The Data Integration Service is an application service that performs data integration tasks such as previewing and running mappings, SQL data services, and web services for the Analyst tool, the Developer tool, and external clients.

The Data Integration Service retrieves metadata from the Model repository when you run a mapping or workflow. The Analyst tool and Developer tool connect to the Data Integration Service to run profile jobs and store profile results in the profiling warehouse.

A Data Integration Service grid is a group of nodes that run Data Integration Service jobs. The Data Integration Service distributes processing across the nodes. For example, when you run a profiling job on a grid, the Data Integration Service splits the work into multiple jobs and assigns each job to a node in the grid.

### *High availability for the Data Integration Service*

The Data Integration Service has the following high availability features:

**Restart and failover**

When a Data Integration Service process becomes unavailable, the Service Manager tries to restart the process or fails the process over to another node based on the service configuration.

**Workflow recovery**

When a Data Integration Service process shuts down unexpectedly, the Data Integration Service can automatically recover canceled workflow instances.

**Data engineering recovery**

If a Data Integration Service process or node fails unexpectedly, the Data Integration Service can recover jobs running on the Spark engine.

#### Data Integration Service grid

When you use a Data Integration Service grid, configure a different Data Integration Service on each node in the grid to allow load distribution and failover. The grid leverages the processing capabilities of all nodes at the same time.

Each grid can have one profiling warehouse. A profiling warehouse stores profiling and scorecard results for profiling and data discovery jobs.

To enable fault tolerance for the profile warehouse, complete the following tasks:

- Replicate the profiling warehouse.
- Set up a load balancer to store metadata in a backup database in case the primary database fails.

## Restart and failover

When a Data Integration Service process becomes unavailable, the Service Manager restarts the Data Integration Service process on the same node or on a backup node.

The restart and failover behavior depends on the following ways that you can configure the Data Integration Service:

**Single node**

When the Data Integration Service runs on a single node and the service process shuts down unexpectedly, the Service Manager tries to restart the service process. If the Service Manager cannot restart the process, the process stops or fails.

**Primary and backup nodes**

When the Data Integration Service runs on primary and backup nodes and the service process shuts down unexpectedly, the Service Manager tries to restart the service process. If the Service Manager cannot restart the process, the Service Manager fails the service process over to a backup node.

A Data Integration Service process fails over to a backup node in the following situations:

- The Data Integration Service process fails and the primary node is not available.
- The Data Integration Service process is running on a node that fails.

**Grid**

When the Data Integration Service runs on a grid, the restart and failover behavior depends on whether the master or worker service process becomes unavailable.

If the master service process shuts down unexpectedly, the Service Manager tries to restart the process. If the Service Manager cannot restart the process, the Service Manager elects another node to run the master service process. The remaining worker service processes register themselves with the new master. The master service process then reconfigures the grid to run on one less node.

If a worker service process shuts down unexpectedly, the Service Manager tries to restart the process. If the Service Manager cannot restart the process, the master service process reconfigures the grid to run on one less node.

The Service Manager restarts the Data Integration Service process based on domain property values set for the amount of time spent trying to restart the service and the maximum number of attempts to try within the restart period.

The Data Integration Service clients are resilient to temporary connection failures during restart and failover of the service.

## Workflow recovery

The Data Integration Service can recover some workflows that are enabled for recovery. Workflow recovery is the completion of a workflow instance from the point of interruption.

A workflow can be interrupted in the following situations:

- An error occurs.
- You cancel the workflow instance.
- You restart the Data Integration Service.
- A Data Integration Service process shuts down unexpectedly.

If you abort the workflow instance, the instance is not recoverable.

The Data Integration Service performs workflow recovery based on the state of the workflow tasks, the values of the workflow variables and parameters during the interrupted workflow instance, and whether the recovery is manual or automatic.

When you configure the workflow options, you can configure the workflow for automatic recovery or manual recovery:

**Automatic recovery**

You can configure automatic recovery of workflow instances.

If you enable a workflow for automatic recovery, the Data Integration Service automatically recovers the workflow from the point of interruption when the Data Integration Service restarts.

If the Data Integration Service runs on a grid and the master service process fails over, all nodes retrieve object state information from the Model repository. The new master automatically recovers workflow instances that were running during the failover and that are configured for automatic recovery.

The Data Integration Service does not recover workflows that are not configured for automatic recovery. You can manually recover these workflows if they are enabled for recovery.

Additionally, the Data Integration Service does not recover any SQL data service, web service, mapping, profile, or preview job that was running during the failover. You must manually restart these jobs.

**Manual recovery**

If you configure manual recovery, you must manually restart the workflow and wait until after the Data Integration Service restarts, if necessary.

### *Data engineering recovery*

If a Data Integration Service process or node fails unexpectedly, the Data Integration Service can recover jobs running on the Spark engine. The Data Integration Service sends the job to another node where job resumes from the point at which the previous node failed. Recovery occurs upon node startup.

If the Data Integration Service runs on a single node, it attempts job recovery when the node is restored. If the Data Integration Service runs on a grid or multiple nodes, when a node fails, the Service Manager fails over to another node.

To use data engineering recovery, you must configure jobs to run on the Spark engine. Configure data engineering recovery in the Data Integration Service properties, and submit the job from the infacmd client.

### *Disaster recovery for the Data Integration Service*

Prepare for disaster recovery by configuring a backup Data Integration Service.

Complete the following tasks:

- Configure the Data Integration Service with a backup on another node or grid.
- Set up a shared file system between the two nodes so that the backup node can recover jobs.
- Replicate the profiling warehouse database.
- Set up a load balancer to store metadata in a backup profiling warehouse database in case the primary database fails.

## Metadata Access Service

The Metadata Access Service is an application service that allows the Developer tool to import and preview metadata from a Hadoop cluster.

The Metadata Access Service contains information about the Service Principal Name (SPN) and keytab information if the Hadoop cluster uses Kerberos authentication. You can create one or more Metadata Access Services on a node.

## *High availability for the Metadata Access Service*

High availability for the Metadata Access Service minimizes interruptions while fetching metadata at design time and enables the domain to react to network failures and failures of the Metadata Access Service.

When a Metadata Access Service process becomes unavailable, the Service Manager tries to restart the process or fails the process over to another node based on the service configuration.

The restart and failover behavior depends on the following ways that you can configure the Metadata Access Service:

**Single node**

When the Metadata Access Service runs on a single node and the service process shuts down unexpectedly, the Service Manager tries to restart the service process. If the Service Manager cannot restart the process, the process stops or fails.

**Primary and backup nodes**

When the Metadata Access Service runs on primary and backup nodes and the service process shuts down unexpectedly, the Service Manager tries to restart the service process. If the Service Manager cannot restart the process, the Service Manager fails the service process over to a backup node.

A Metadata Access Service process fails over to a backup node in the following situations:

- The Metadata Access Service process fails and the primary node is not available.
- The Metadata Access Service process is running on a node that fails.

The Service Manager restarts the Metadata Access Service process based on domain property values set for the amount of time spent trying to restart the service and the maximum number of attempts to try within the restart period.

The Metadata Access Service clients are resilient to temporary connection failures during restart and failover of the service.

## *Disaster recovery for the Metadata Access Service*

Prepare for disaster recovery by configuring the Metadata Access Service with a backup on another node.

# Analyst Service

The Analyst Service is an application service that runs the Analyst tool and manages the connections between the service components and the users who log in to the Analyst tool.

The Analyst Service connects to the following Informatica services and tools:

- Data Integration Service to run profiles, scorecards, mapping specifications, and workflows.
- Model Repository Service to identify a Model repository.
- Metadata Manager Service for data lineage analysis on scorecards in the Analyst tool.
- Search Service to enable and manage searches in the Analyst tool.

Additionally, the Analyst Service connects to a flat file cache directory to store uploaded flat files and a business glossary export file directory.

## *High availability for the Analyst Service*

To use high availability, complete the following tasks for the Analyst Service:

1. Set up a shared file system between Analyst Services to maintain the flat file cache directory.
2. Set up a load balancer for the Analyst tool web interface across the Analyst Services.

When a service failure occurs, change the node on which the Analyst Service is available to start the service again.

### *Disaster recovery for the Analyst Service*

The Analyst Service disaster recovery strategy relies on the Model Repository Service. The Model repository database stores the objects that the Analyst Service references.

To recover the Analyst Service during a disaster, the Model Repository Service must be running in the new region or availability zone. To start the Analyst Service, use the node that the Model Repository Service is running on and assign the node to the Analyst Service to start the Analyst Service on the same node.

## Content Management Service

The Content Management Service is an application service that manages reference data. It provides reference data information to the Data Integration Service, Developer tool, and Analyst tool.

A master Content Management Service maintains probabilistic model and classifier model data files across the domain. When you compile a probabilistic model or classifier model in the Developer tool, you update the model files on the master Content Management Service machine.

If a node in the domain runs a Content Management Service, the node stores local copies of the probabilistic model and classifier model files. The master Content Management Service synchronizes the probabilistic model and classifier model files on the domain nodes with the master Content Management Service files every 10 minutes.

### *High availability for the Content Management Service*

To use high availability, configure a Content Management Service on a backup node.

If you use user-generated content, such as content sets that contain probabilistic or classifier models, verify that the Content Management Service on the backup node runs continually. When the Content Management Service on the backup node runs continually, any change to the user-generated content on the primary node is copied to the backup node.

### *Disaster recovery for the Content Management Service*

Prepare for disaster recovery by configuring a backup Content Management Service.

Complete the following tasks:

- Configure the Content Management Service with a backup on another node.
- Replicate the reference database.
- Set up a load balancer to store metadata in a backup database in case the primary database fails.

## Search Service

The Search Service is an application service that manages search in the Analyst tool and Business Glossary Desktop.

The Search Service performs each search on a search index instead of the Model repository or profiling warehouse. To create the search index, the Search Service extracts information about content from the Model repository and profiling warehouse. You can configure the interval at which the Search Service extracts this information.

The Search Service can fetch results for data objects such as mapping specifications, profiles, reference tables, rules, and scorecards. It can also return the following additional results:

- Related assets

- Business terms
- Policies
- Profile results from a profiling warehouse
- Domain discovery results from a profiling warehouse

You can associate each Search Service with one Model repository and one profiling warehouse. To perform searches on multiple Model repositories or profiling warehouses, you must create multiple Search Services.

### *High availability for the Search Service*

To use high availability, create another Search Service on a backup node. When a service failure occurs, manually start the Search Service on the running backup node.

### *Disaster recovery for the Search Service*

Create another Search Service on a backup node in a different region or availability zone. When a disaster occurs, manually start the Search Service on the running backup node.

During disaster recovery, the Search Service relies on the Model Repository Service. The Search Service creates indexes on the information that it extracts from the contents of data objects stored in the Model repository and profiling warehouse.

To recover the Search Service during a disaster, the Model Repository Service must be running in the new region or availability zone. Therefore, you can create the backup Search Service on the backup node for the Model Repository Service.

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